William C. Siemens

| Berkeley, CA | (530) 758-6264 | w.siemens04@gmail.com |
| www.linkedin.com/in/will-siemens | Project Portfolio |

Education

University of California, Berkeley — **May 2027**
Masters of Science in Mechanical Engineering — GPA: N/A/4.0

University of California, Berkeley — **May 2026**
Bachelor of Science in Mechanical Engineering — GPA: 3.960/4.0

Experience

Mechanical Engineering Expert - *Mercor*, Remote — **August 2026 - Present**

- Assessed AI-generated mechanical engineering content for technical accuracy, leveraging professional experience to provide well-reasoned feedback and curate essential reference materials.
- Collaborated with cross-functional subject-matter experts to share industry insights regarding engineering workflows, tools, and best practices.

Undergraduate Research, *Kamrin Group* - Advisor: Professor Kenneth Kamrin — **January 2025 - Present**

- Co-Leading an independent project to optimize the open-source PeGS (Photoelastic Grain Solver) codebase, which estimates inter-particle forces in 2D granular flow from photoelastic fringe patterns.
- Replaced legacy models with complex energy potential methods, improving accuracy and computational speed by two orders of magnitude.
- Refactored codebase for vectorization and parallelization; implemented custom Jacobian-based solvers for faster particle solution.
- Working closely with one research partner and faculty advisor to improve solver scalability and robustness.

Design Intern, *Saturday Robotics*, Boston, MA — **May 2023 - July 2024**

- Led engineering design projects under an advisors guidance to enhance robotic systems and operations at a stealth-mode industrial robotics startup, focused on process automation and environmental solutions.
- Led concept design and development of large-scale steel superstructures optimized for spatial efficiency and integration of industrial machinery.
- Engineered versatile bent sheet metal adapter plates, enabling five unique mounting orientations to reduce part count and improve cable management and ergonomics of electronic integration.
- Designed and manufactured custom fit-tools to support specialized assembly and maintenance needs.

Course Reader - *Department of mechanical engineering*, UC Berkeley. — **Fall 2024, 2025, & Spring 2026 Semesters**

- Class grader for undergraduate mechanical engineering course.
- Delivered detailed feedback while maintaining grading accuracy, fairness, and deadlines.

Highlighted Projects

Card Shuffler and Dealer, Course: Mechatronics Design — **Spring 2026**

- Designed and manufactured a compact, fully integrated mechatronics platform using SolidWorks and 3D printing to automate card shuffling, alignment, and multi-player distribution.
- Engineered a custom mechanical transmission system utilizing synchronized gear trains, belt drives, and a friction-based roller concept to separate single cards from a stack while preventing double-feeding and jams.
- Developed embedded control logic on an ESP32 microcontroller to manage multi-axis motion control (stepper, servo, and DC motors), coordinate software timing with mechanical tolerances, and interface with wireless player button assemblies.
- Implemented automated calibration and homing routines through iterative physical testing and limit-switch integration, ensuring a repeatable card ejection trajectory across varied player positions.

Speaker Remote, Course: Electronics & Internet of Things — **Spring 2025**

- Engineered a wireless ESP32-based remote for Audioengine A5+ speakers, integrating it with existing hardware to control volume and mute via IoT without line-of-sight.
- Wrote and implemented software with deep sleep functionality, extending battery life and health.
- Developed precise encoder signal interception and logic-level shifting for seamless volume synchronization and input queuing.
- Created sleek, ergonomic 3D-printed housing with USB-C charging and integrated alphanumeric display for real-time feedback.

Awards and Extracurricular Activities

UC Berkeley Varsity Track and Field Athlete — **August 2022 - May 2027**
Guangyi R. Zheng Prize — **Undergraduate Graduation 2026**
Honors to Date — **All But 2 Semesters**
Dean's List - College of Engineering — **Spring 2023, Fall 2023, Spring 2024**